Formula Sheet

Slope-intercept form	Slope formula	Point-Slope formula
$y = mx + b$	$m = \dfrac{y_2 - y_1}{x_2 - x_1}$	$y - y_1 = m(x - x_1)$

P =principal/present value r =annual interest rate m =compoundings per year $n = tm$
A =future/maturity value t =number of years $i = {}^{r}/_{m}$ =rate per period r_E =effective rate

Simple Interest	Compound Interest	Continuous Compounding
$A = P(1 + rt)$	$A = P(1 + i)^n$	$A = Pe^{rt}$
$P = \dfrac{A}{1 + rt}$	$P = A(1 + i)^{-n}$	$P = Ae^{-rt}$
	$r_E = \left(1 + \dfrac{r}{m}\right)^m - 1$	$r_E = e^r - 1$

R =periodic payment P =present value of annuity n =number of periods
S =future value i =rate per period

Future Value of an Annuity	Sinking Fund Payment
$S = R\left[\dfrac{(1+i)^n - 1}{i}\right]$	$R = \dfrac{S \cdot i}{(1+i)^n - 1}$

Present Value of an Annuity	Amortization Payments
$P = R\left[\dfrac{1 - (1+i)^{-n}}{i}\right]$	$R = \dfrac{P}{[1 - (1+i)^{-n}]}$

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N	:	number of payments
`I%`	:	interest rate
`PV`	:	present value
`PMT`	:	payment size
`FV`	:	future value
`P/Y`	:	payments per year
`C/Y`	:	compoundings per year